

Batch-Tiled Evaluation Overstates the Cost of Causal Sleep Staging: Evidence from Two Datasets

Jahnvi R, Swapnil S, Shreeya S Methuku, Shreshtha Dixit

Department of Computer Science (AIML) and Engineering, PES University, Bengaluru, India

Abstract—A sleep-staging model intended for a wearable device may use only past signal, and this constraint is generally assumed to cost accuracy. We measure that cost with a matched ablation in which one architecture is trained with and without access to future signal, holding parameter count, data, folds and seeds fixed, and run the identical experiment on two datasets. Under the batch-tiled protocol standard in this literature, which covers a recording with non-overlapping context windows, the causal constraint costs 0.025 Cohen’s κ on 78 Sleep-EDF subjects and 0.033 on the 25 subjects of DOD-H, and on both appears concentrated in REM. We show that most of this is produced by the protocol. Tiling scores predictions at every position of the context buffer, including positions with almost no accumulated history, which a streaming system passes through once at start-up; the causal model is crippled there while the non-causal model is not, because it retains lookahead. Moving the same weights to a streaming protocol therefore lifts the two arms unequally, and the excess lift to the causal arm is $+0.0314$ and $+0.0292 \kappa$, positive in all 30 paired folds across the two datasets, with Nadeau–Bengio corrected $t(14) = 12.69$ and 6.48 . We verify the mechanism arithmetically: the tiled score is the mean of a curve over buffer positions, reproduced to within 0.003κ in all four model–dataset combinations, and the penalty is generated by positions carrying under 30 seconds of history. Re-measured under the deployment protocol the causality cost is $+0.0065$ and -0.0037κ , neither resolvable, and the apparent REM concentration falls from 0.060 F1 to 0.002 and 0.023. Reported causality penalties in this literature should be read together with the evaluation protocol that produced them.

Index Terms—sleep staging, EEG, causal models, real-time inference, evaluation methodology

I. INTRODUCTION

Most automated sleep-staging models classify 30-second epochs and are free to use signal from later in the recording, or the whole night, when labelling an earlier epoch [1], [2]. That is appropriate for offline analysis of a completed study but unsuitable for a wearable device, which must report a stage as it occurs. Causality is therefore a real design constraint, and the natural question is what it costs.

That question is rarely answered cleanly. Published causal and non-causal models differ in capacity, data, preprocessing and optimization simultaneously, so the gap between them is not attributable to causality in particular. High output resolution is not the open problem either: U-Sleep [3] and its predecessor U-Time [15] have emitted up to 7,680 stages per minute since 2021, using a bidirectional model that reads several minutes of future signal. What has not been measured is what that resolution costs when the model may not look forward at all.

Causal staging itself is an active direction. Bresch et al. [13] apply recurrent networks to real-time single-channel staging, CausalAttenNet [12] combines causal convolution with long-term attention, and Micro SleepNet [14] targets wearable hardware. We claim no priority over these on the architecture. To our knowledge, however, none reports a matched ablation in which the identical network is trained with and without access to future signal on identical folds and seeds, which is what licenses attributing a gap to causality rather than to capacity, and none reports how the answer moves with the evaluation protocol.

We answer it with a matched ablation: one architecture, trained twice, differing only in whether convolutions are left-padded and attention is masked, run end to end on two datasets. The first result is the expected one. Under the evaluation protocol this literature normally uses, causality costs 0.025κ on Sleep-EDF-78 and 0.033 on DOD-H. The second result is that most of that number is produced by the protocol rather than by the constraint, and this paper is about the second result.

Our contribution is one finding with three consequences. The finding is that batch-tiled evaluation systematically penalizes causal models, measured as a within-arm quantity that does not require the two arms to be comparable, replicated on two datasets that differ in montage, sampling rate, scoring team, class balance and subject count. The consequences are that the reported causality penalty is largely an artifact, that its apparent concentration in REM is the same artifact, and that a corpus too small to resolve the between-arm ablation can still resolve this effect decisively, which is what makes the measurement worth adopting.

II. METHOD

A. Datasets

Sleep-EDF-78 [4], [5]: 78 subjects, 153 recordings, single channel Fpz-Cz at 100 Hz, 1,629 hours after trimming each night to 30 minutes of Wake either side of the sleep period. **DOD-H** [6]: 25 subjects, 205.5 hours, channel F3-M2 resampled from 250 to 100 Hz with a polyphase anti-aliasing filter, no Wake trimming, since Wake is already 12.3% of the corpus against 68% in untrimmed Sleep-EDF. Both are scored to five AASM classes [7] in 30-second epochs; each epoch label is replicated across its 30 one-second targets, so the network is supervised at 1 Hz. Both use subject-wise 5-fold cross-validation with no subject crossing the train/test boundary. Nothing about the architecture, the training recipe

or the two protocols changes between datasets; the focal-loss class weights are recomputed per training fold, so the differing class balances are handled automatically.

B. Model and the Causality Ablation

A multi-resolution causal CNN (kernels of 50 and 400 samples, left-aligned pooling to 1 Hz) feeds three residual blocks of causal dilated convolutions [8] (dilations 1, 2, 4, 32 channels) and a four-head self-attention layer [9] with a causal mask, then a linear classifier emitting one five-class prediction per second. Training uses focal loss [10] with $\gamma = 2$ and per-fold class weights, early stopping on validation κ . Total 30,757 parameters; 0.026 ms per second of EEG on one CPU thread.

Causality is enforced in exactly two places: convolutions pad only the left by $(K - 1)d$, and the attention mask sets $M_{ij} = -\infty$ for $j > i$. The non-causal arm removes both and changes nothing else, so the two arms are identical in capacity and differ only in access to future signal.

Causality also has to hold upstream of the network. Per-epoch z -scoring, near-universal in this literature, normalizes an early sample using statistics over the whole epoch including its future. We replace it with a trailing 30-second statistic computable one sample at a time. Closing this leak raises κ from 0.6406 to 0.6649 (causal) and 0.6693 to 0.6898 (non-causal), so it is a modelling gain and never a trade. The full path from raw sample to logit is verified causal by perturbation test, and a streaming front end reproduces the offline labels exactly over a 79,440-second held-out recording. Because the two corpora are recorded in different amplitude units, we also verify that the normalizer removes a $1000\times$ change in input gain to within 5×10^{-7} .

C. Evaluation Protocols

Tiled. The recording is covered by non-overlapping context windows and every position of every window contributes one scored prediction. This is the default in every implementation we have examined, including our own earlier work. Each window begins from an empty history.

Streaming. The model recomputes every 30 s against a full buffer and only the freshest 30 predictions are kept, so after the opening window every scored second carries at least 90 seconds of genuine history. This is what a deployed system does. Every held-out second is scored exactly once under each protocol, from the same trained weights.

D. Statistics

Each arm is trained under three seeds $\times$ five folds per dataset, giving 15 paired measurements and 60 trained models in total. Because repeats reuse subjects, the primary test is the Nadeau–Bengio corrected resampled t -test [11], which inflates the variance estimate by $1/n + 1/(k - 1) = 1/15 + 1/4$ here. Bootstrap intervals resample whole subjects. Pairing is exact: within a seed the two arms use byte-identical subject lists.

TABLE I
COST OF CAUSALITY UNDER EACH PROTOCOL. 15 PAIRED FOLDS PER ROW, NADEAU–BENGIO CORRECTED

Dataset	Protocol	Causal	Non-c.	Δ	Loses	p
Sleep-EDF-78	Tiled	0.6649	0.6898	−0.0249	14/15	0.005
Sleep-EDF-78	Streaming	0.6998	0.6934	+ 0.0065	7/15	0.47
DOD-H	Tiled	0.6329	0.6659	−0.0329	10/15	0.22
DOD-H	Streaming	0.6674	0.6711	− 0.0037	6/15	0.89

TABLE II
PROTOCOL GAIN BY ARM: $\kappa(\text{STREAMING}) - \kappa(\text{TILED})$, PAIRED WITHIN FOLD

	Sleep-EDF-78	DOD-H
Causal arm gains	+0.0350	+0.0345
Non-causal arm gains	+0.0036	+0.0052
Excess to the causal arm	+0.0314	+0.0292
Folds with positive excess	15/15	15/15
Bootstrap 95% CI	[.0291, .0334]	[.0253, .0332]
Corrected $t(14)$	12.69	6.48
p	4.6×10^{-9}	1.4×10^{-5}
Per-seed range	.0310–.0319	.0285–.0297

III. RESULTS

A. The Protocol Change Is Worth Ten Times More to the Causal Arm

Table I gives the causality cost under each protocol. On Sleep-EDF-78 the tiled penalty is resolved and the streaming one is not. On DOD-H neither is: with five subjects per held-out fold the corrected test returns $p = 0.22$ on a point estimate larger than Sleep-EDF-78’s. A claim built on this table alone would therefore rest on one dataset.

The decisive statistic is a different one, and it is decisive because it does not require the two arms to be comparable. Ask instead how much each arm gains from the protocol change, which is a within-model difference on identical weights and identical held-out seconds (Table II).

The protocol change is worth roughly ten times as much to the causal arm as to the non-causal arm; the excess is positive in every one of the thirty paired folds across the two datasets, the smallest single-fold value anywhere being +0.0173; and two corpora differing in electrode montage, sampling rate, scoring team, class balance, subject count and recording length agree on its size to within 0.002 κ . The aggregate cost follows by arithmetic rather than as an independent observation: a tiled penalty of −0.0249 and −0.0329, an excess of +0.0314 and +0.0292, and a streaming residual of +0.0065 and −0.0037. The protocol accounts for essentially all of the measured penalty on Sleep-EDF-78 and about 89% of it on DOD-H.

B. The Mechanism, Verified

Define permitted latency L : the label for second t is emitted at $t + L$, so the model sees L seconds of future and $L_{\text{buf}} - L$ of past, and $L = 0$ is real-time operation. One forward pass over a window yields a prediction at every position, and position p

TABLE III
 κ BY BUFFER QUARTILE, SEED 42, 120 s BUFFER. POSITIVE Δ MEANS THE CAUSAL ARM PAYS

History (s)	Sleep-EDF-78			DOD-H		
	Caus.	Non-c.	Δ	Caus.	Non-c.	Δ
91–120	0.6982	0.6930	−0.0052	0.6904	0.6961	+0.0057
61–90	0.6923	0.6969	+0.0046	0.6887	0.7044	+0.0157
31–60	0.6753	0.6911	+0.0158	0.6737	0.6970	+0.0233
1–30	0.5847	0.6752	+0.0905	0.5622	0.6685	+0.1063

carries exactly $L_{\text{buf}} - 1 - p$ seconds of lookahead, so a single sweep measures every latency.

Averaging the resulting curve over all 120 buffer positions reproduces that model’s tiled score to within 0.003 κ in all four model–dataset combinations: 0.6626 against a measured 0.6634 and 0.6890 against 0.6912 on Sleep-EDF-78, 0.6537 against 0.6567 and 0.6915 against 0.6922 on DOD-H. The tiled protocol is, arithmetically, this average.

Table III locates the penalty, and the two datasets show the same shape. At buffer positions carrying 91–120 seconds of history the arms are level. Across the last quartile the causal arm loses 0.114 and 0.128 κ against its own best quartile while the non-causal arm loses 0.018 and 0.028; that gap is the artifact, and a streaming system occupies those positions once, at start-up.

Fig. 1 gives the full curves. The causal arm declines monotonically from $L = 0$: it cannot read future signal, so larger L only costs it history, and at $L = 119$ it collapses to $\kappa = 0.317$ and 0.252. The non-causal arm traces an inverted U and peaks at $L = 38$ s on Sleep-EDF-78 and $L = 46$ s on DOD-H.

At each arm’s best operating point the two datasets disagree, and we report that rather than the agreement because it bounds the claim. On Sleep-EDF-78 the causal arm reaches $\kappa = 0.7008$ at zero latency against a non-causal optimum of 0.6977 at 38 s of delay, a margin of 0.003 κ we would not defend as more than a tie. On DOD-H the causal arm reaches 0.6922 at zero latency and the non-causal optimum is 0.7058 at 46 s, exceeding it by 0.0136. What replicates across the two datasets is the protocol asymmetry; the stronger statement that a non-causal model cannot beat a causal one at any latency holds on Sleep-EDF-78 and fails on DOD-H, where roughly 0.014 κ is available in exchange for 46 seconds of delay.

C. The REM Concentration Is the Same Artifact

Under tiling, REM appears to pay 0.060 F1 on both datasets, four times the next-largest class on Sleep-EDF-78 and the only class whose penalty there survives Holm correction across the five per-class tests. Under the deployment protocol the difference is +0.002 and +0.023, and N1, N2 and N3 invert on both datasets, the causal arm ending ahead on all three (Table IV). This is mechanistically sensible: REM is defined partly by surrounding context, which is why a technician resolves it against neighbouring epochs, so it is the class most damaged by a nearly empty buffer. Two residuals remain rather than vanishing. REM still costs 0.023 F1 on DOD-H at

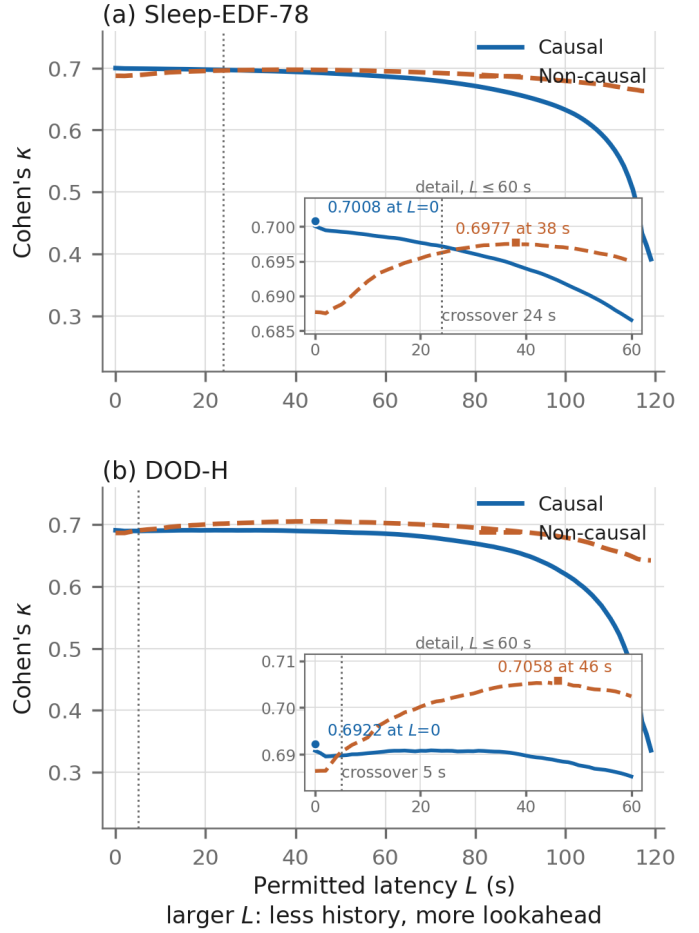


Fig. 1. Cohen’s κ against permitted latency L , both arms, both datasets, 5-point moving average over L . Larger L means less history and more lookahead within a fixed 120 s buffer. The causal collapse at large L is the cold-start behaviour the tiled protocol averages into its score.

TABLE IV
 PER-CLASS F1 DIFFERENCE BY PROTOCOL, 15 PAIRED FOLDS PER COLUMN. POSITIVE MEANS THE CAUSAL ARM PAYS

Class	Sleep-EDF-78		DOD-H	
	Tiled	Streaming	Tiled	Streaming
W	+0.018	+0.006	+0.046	+0.035
N1	+0.025	−0.009	+0.007	−0.018
N2	+0.002	−0.017	+0.014	−0.004
N3	+0.008	−0.020	+0.015	−0.004
REM	+0.060	+0.002	+0.060	+0.023
Macro	+0.023	−0.007	+0.028	+0.006

deployment, and Wake costs 0.035 there under both protocols; DOD-H is not Wake-trimmed, so its Wake is sleep-adjacent and transitional throughout, which is exactly the case where seeing forward helps.

D. What the Per-Second Rate Buys

A per-second rate is only useful if it lets a system report something sooner. Charging both systems for their evidence, a change counts as reported once the new stage has been held

for h seconds, while an epoch-level system cannot emit the label for epoch k until epoch k has ended. At $h = 10$ s the per-second system reports first in 94.8% of transitions on Sleep-EDF-78 and 95.6% on DOD-H, with a median 19 seconds saved on both. The advantage is conditional on h and we report the full sweep: it falls to 66.5% at $h = 20$ and 29.6% at $h = 30$, where the epoch system leads, because requiring 30 seconds of confirmation reimposes the epoch and discards the resolution.

E. Why the Estimand Matters at This Sample Size

DOD-H's fold-to-fold κ standard deviation is 0.098, four times Sleep-EDF-78's 0.026, because five subjects are held out per fold rather than sixteen. The same fifteen runs cannot resolve the between-arm causality difference ($p = 0.22$) and do resolve the within-arm protocol gain ($p = 1.4 \times 10^{-5}$), because the latter scores identical weights on identical seconds and so removes model variance from the comparison entirely. At the subject counts common in this literature the choice of estimand decides what is measurable, which is the practical reason to report the protocol comparison alongside the ablation.

IV. DISCUSSION AND CONCLUSION

An earlier version of this work reported that the causal constraint costs 0.029κ and that the cost concentrates in REM. Both are reproducible under the protocol that produced them and both are largely artifacts of it. The mechanism is not subtle once seen: tiling a recording with independent context windows repeatedly destroys the accumulated history that defines a streaming model and then scores it on the wreckage, and it does so asymmetrically, since a model that can read forward is largely unaffected at the start of a window and a model that cannot is crippled there.

Three things bound the claim. The streaming protocol denies the non-causal arm the lookahead that defines it, so this compares deployment options rather than architectures; at matched latency the answer is dataset-dependent, with lookahead worth nothing on Sleep-EDF-78 and about 0.014κ at 46 seconds of delay on DOD-H. The non-causal arm is our own matched twin rather than a state-of-the-art offline model, so the comparison bounds what causality costs within this capacity class. And both datasets are single-channel EEG scored to AASM conventions with one architecture; the mechanism argues that any model whose predictions depend on accumulated history should show the effect, but two instances of one architecture is not a general result.

What we would like adopted is smaller than a conclusion about causality and more robust than one. Any matched ablation of a streaming model against a bidirectional one should report both protocols, because on our data the two numbers differ by more than the effect being measured, and because the within-arm comparison is resolvable on corpora where the between-arm one is not. Code, per-fold predictions, splits, and the analysis for both datasets are released so the measurement can be repeated without retraining.

REFERENCES

- [1] A. Supratak, H. Dong, C. Wu, and Y. Guo, "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 11, pp. 1998–2008, 2017.
- [2] E. Eldele et al., "An attention-based deep learning approach for sleep stage classification with single-channel EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 809–818, 2021.
- [3] M. Perslev, S. Darkner, L. Kempfner, M. Nikolic, P. J. Jennum, and C. Igel, "U-Sleep: Resilient high-frequency sleep staging," *npj Digit. Med.*, vol. 4, art. 72, 2021.
- [4] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [5] B. Kemp, A. H. Zwiderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Obery, "Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG," *IEEE Trans. Biomed. Eng.*, vol. 47, no. 9, pp. 1185–1194, 2000.
- [6] A. Guillot, F. Sauvet, E. H. During, and V. Thorey, "Dreem Open Datasets: Multi-scored sleep datasets to compare human and automated sleep staging," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 9, pp. 1955–1965, 2020.
- [7] R. B. Berry et al., *The AASM Manual for the Scoring of Sleep and Associated Events, Version 2.6*. Darien, IL: American Academy of Sleep Medicine, 2020.
- [8] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv:1803.01271*, 2018.
- [9] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 5998–6008.
- [10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2980–2988.
- [11] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Mach. Learn.*, vol. 52, no. 3, pp. 239–281, 2003.
- [12] J. Pan, Y. Feng, P. Zhao, X. Zou, A. Hou, and X. Che, "CausalAttenNet: A fast and long-term-temporal network for automatic sleep staging with single-channel EEG," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2024.
- [13] E. Bresch, U. Großekathöfer, and G. Garcia-Molina, "Recurrent deep neural networks for real-time sleep stage classification from single channel EEG," *Front. Comput. Neurosci.*, vol. 12, art. 85, 2018.
- [14] G. Liu et al., "Micro SleepNet: Efficient deep learning model for mobile terminal real-time sleep staging," *Front. Neurosci.*, vol. 17, art. 1218072, 2023.
- [15] M. Perslev, M. Jensen, S. Darkner, P. J. Jennum, and C. Igel, "U-Time: A fully convolutional network for time series segmentation applied to sleep staging," in *Advances in Neural Information Processing Systems 32*, 2019.